“Counterfactual Explanations”

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Interpretable or Explainable AI

Biran and Cotton 2017, which was also used by T. Miller 2019:

> “Interpretability is the degree to which a human can understand the cause of a decision.”

Another good one is by Kim, Khanna, and Koyejo 2016:

> “A method is interpretable if a user can correctly and efficiently predict the method’s results”

Interpretable or Explainable AI

Interpretability

Mapping an abstract concept from the models into an understandable form.

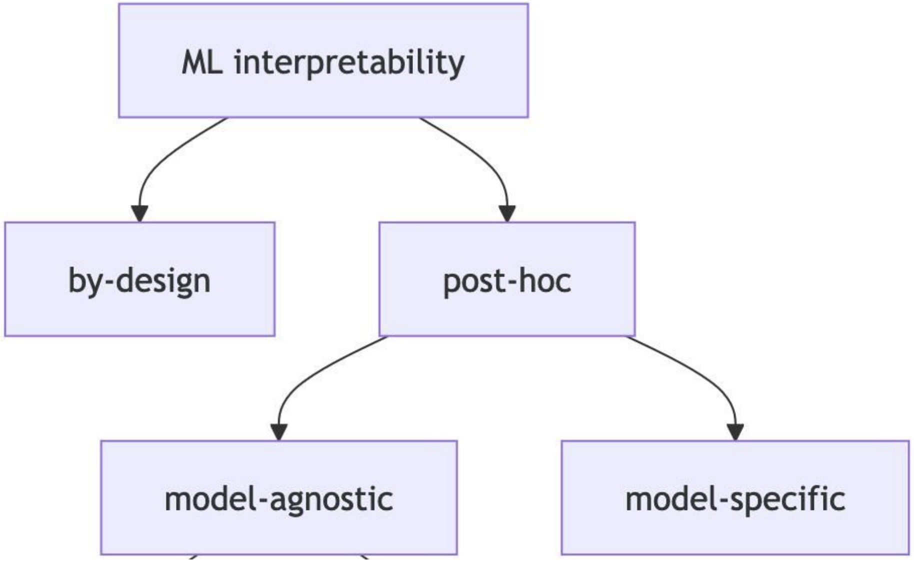
Explainability

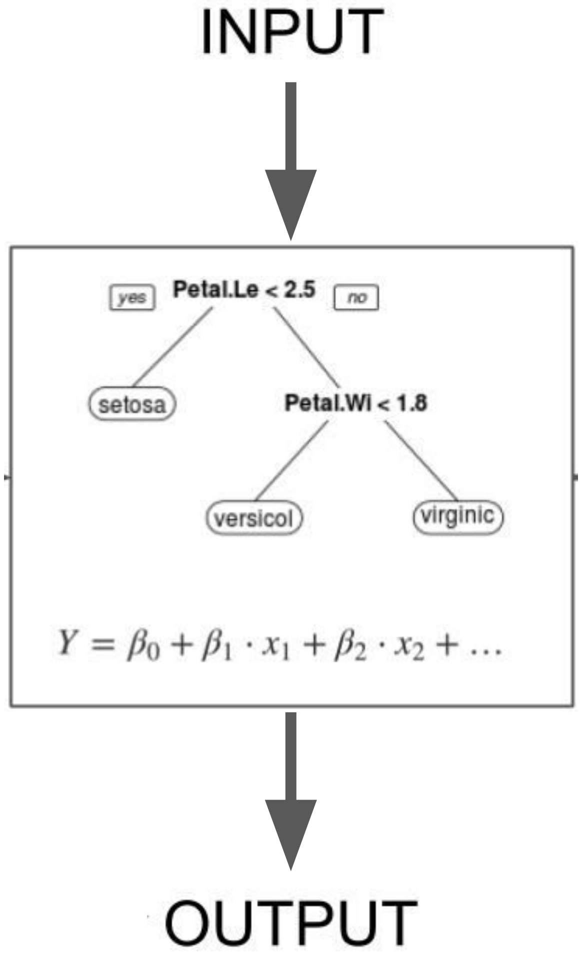
Stronger term requiring interpretability and additional context.

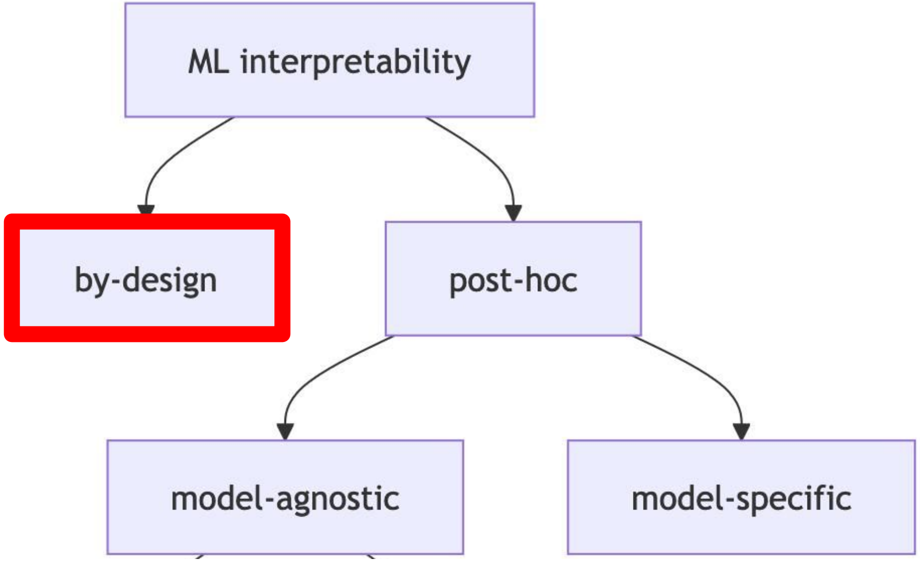
In Practice: people use both interchangeably.

An explanation is the answer to a why-question (Miller 2017).

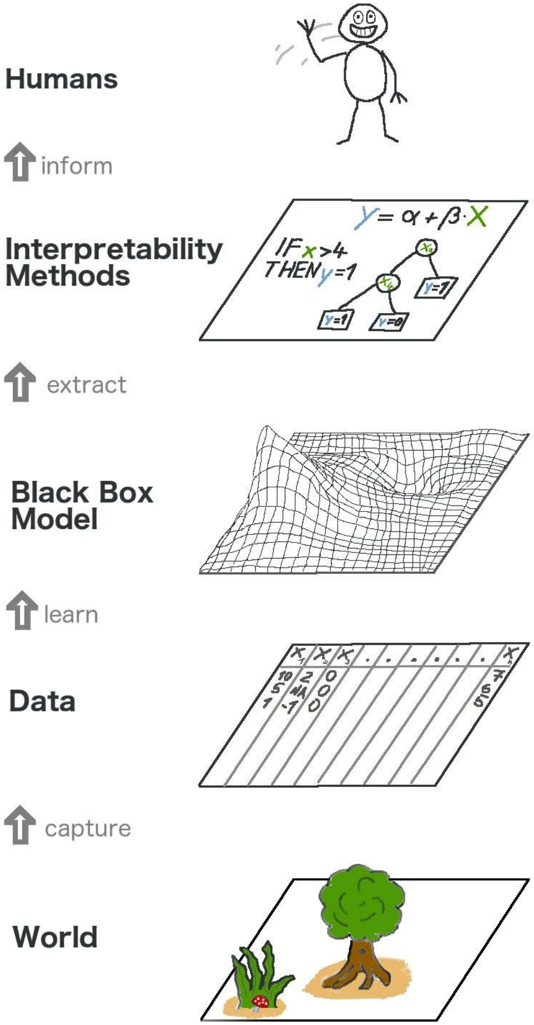
- Why did the treatment not work on the patient?
- Why was my loan rejected?
- Why have we not been contacted by alien life yet?

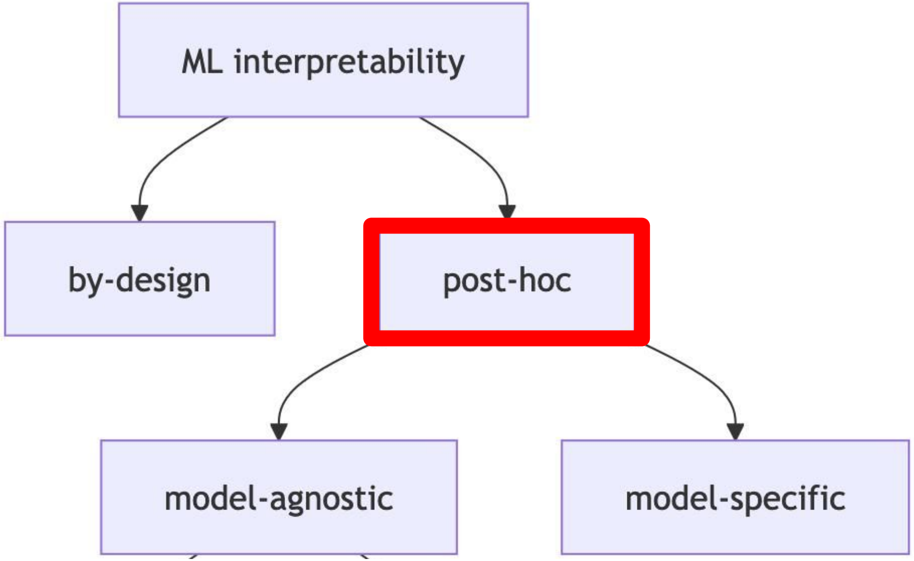

ML interpretability
by-design
post-hoc
model-agnostic
model-specific





- **Linear regression**
- **Logistic regression**
- **Decision trees**


Humans
inform
Interpretability Methods
y = α + β·X
IF x>4 THEN y=1
extract
Black Box Model
learn
Data
capture
World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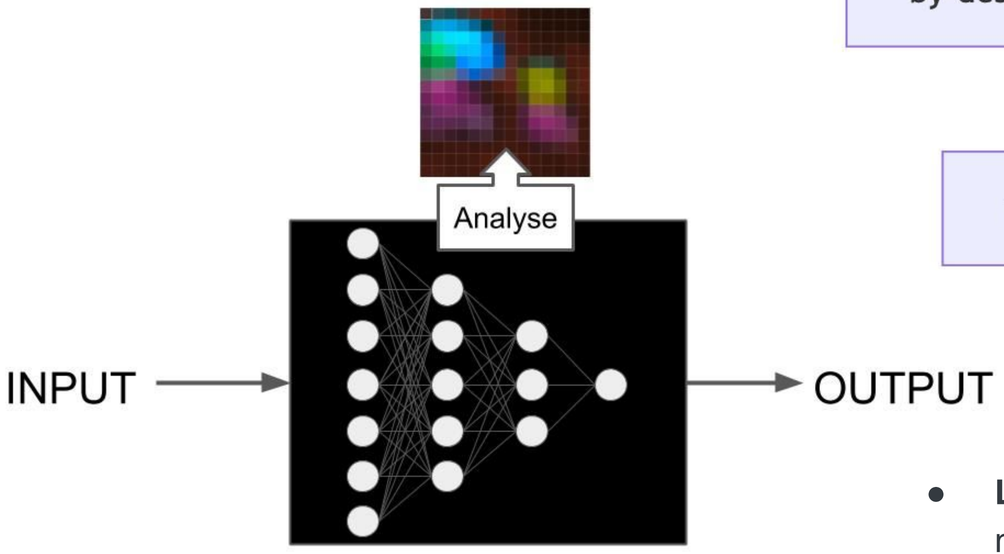
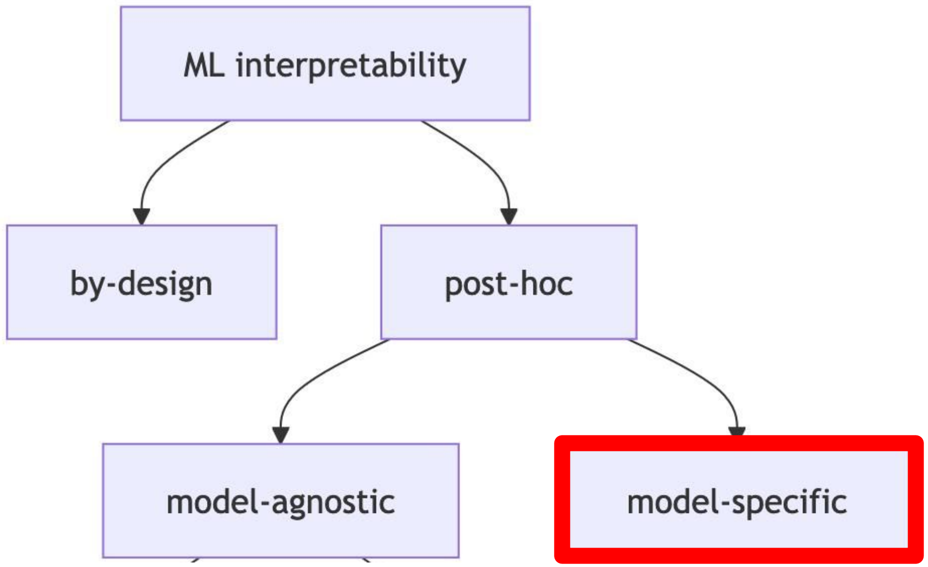
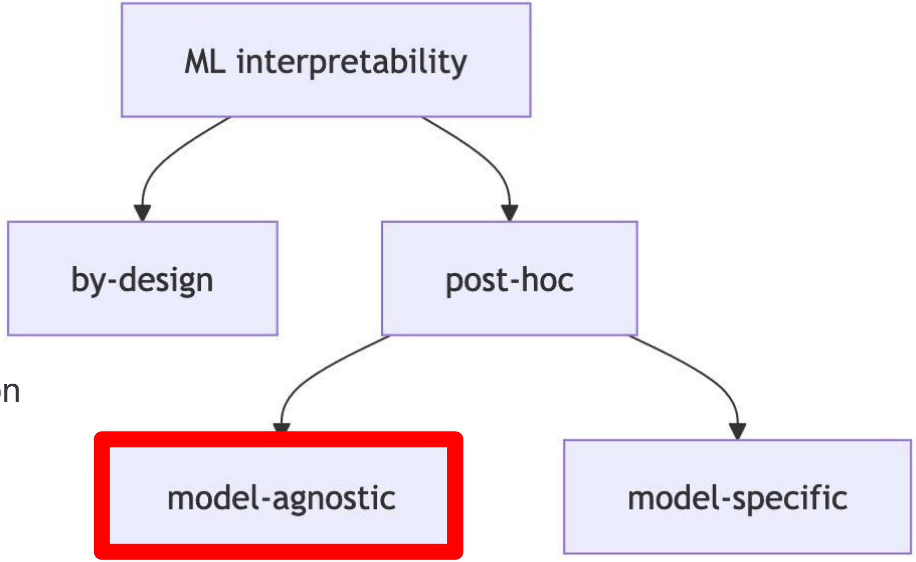
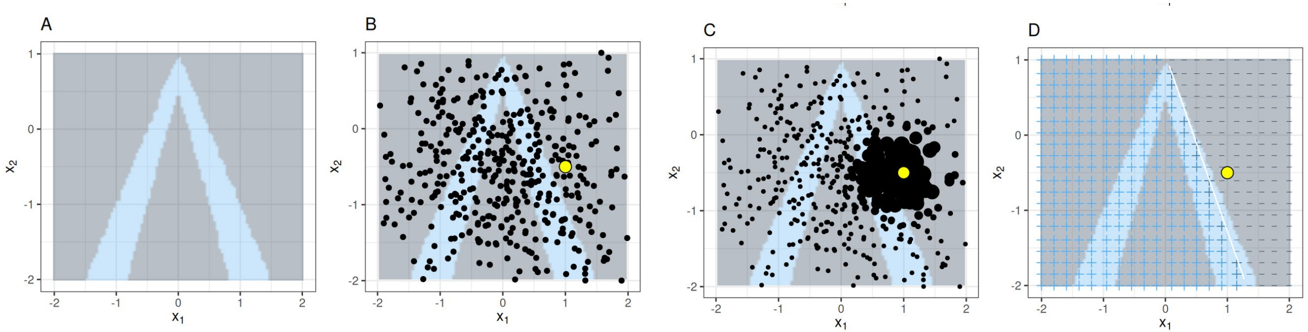




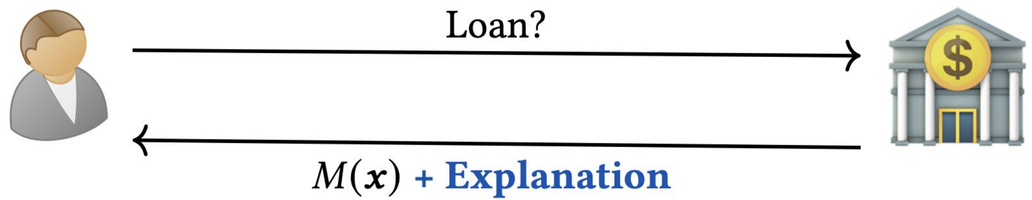
- **Learned Features:** What features did the neural network learn?
- **Saliency Maps:** How did each pixel contribute to a particular prediction?

- **Local surrogate models (LIME)** explain a prediction by replacing the complex model with a locally interpretable model.
- **SHAP/Shapley values** fairly assign the prediction to individual features.
- **Counterfactual explanations!** explain a prediction by examining which features would need to be changed to achieve a desired prediction.





What is Counterfactual Explanations?



Factual $\boldsymbol{x}$

Age	Income	Debt	Accounts
28	800	200	5

M( ) = no loan

Counterfactual $\boldsymbol{x}_{cf}$

Age	Income	Debt	Accounts
28	1000	200	3

M( ) = loan

Explanation: Your loan will be approved,
if you increase income by $200 and close two accounts.

What is Counterfactual Explanations?

- A counterfactual explanation describes a causal situation in the form: **"If X had not occurred, Y would not have occurred"**

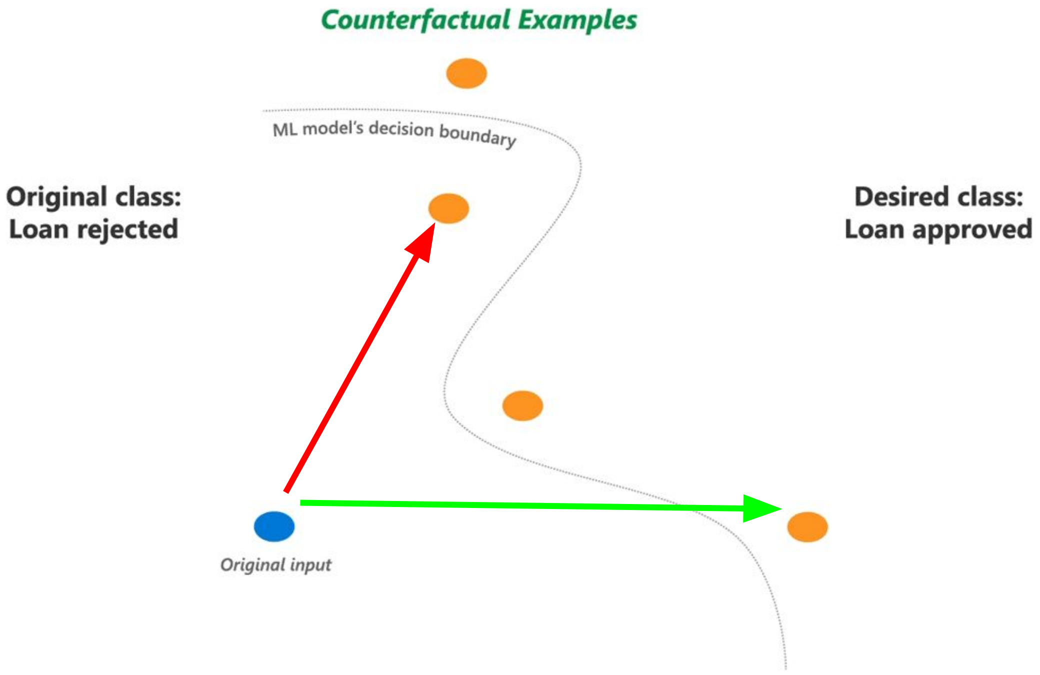
- In interpretable machine learning, counterfactual explanations can be used to **explain predictions of individual instances**

- A counterfactual explanation of a prediction describes **the change to the feature values** that changes the prediction to a predefined output.

- **Counterfactuals are human-friendly explanations**, because they are contrastive to the current instance and because they are selective, meaning they usually focus on a small number of feature changes.

Desired Properties: **Validity**

The counterfactual must produce the desired prediction change.



Metric: **Validity**

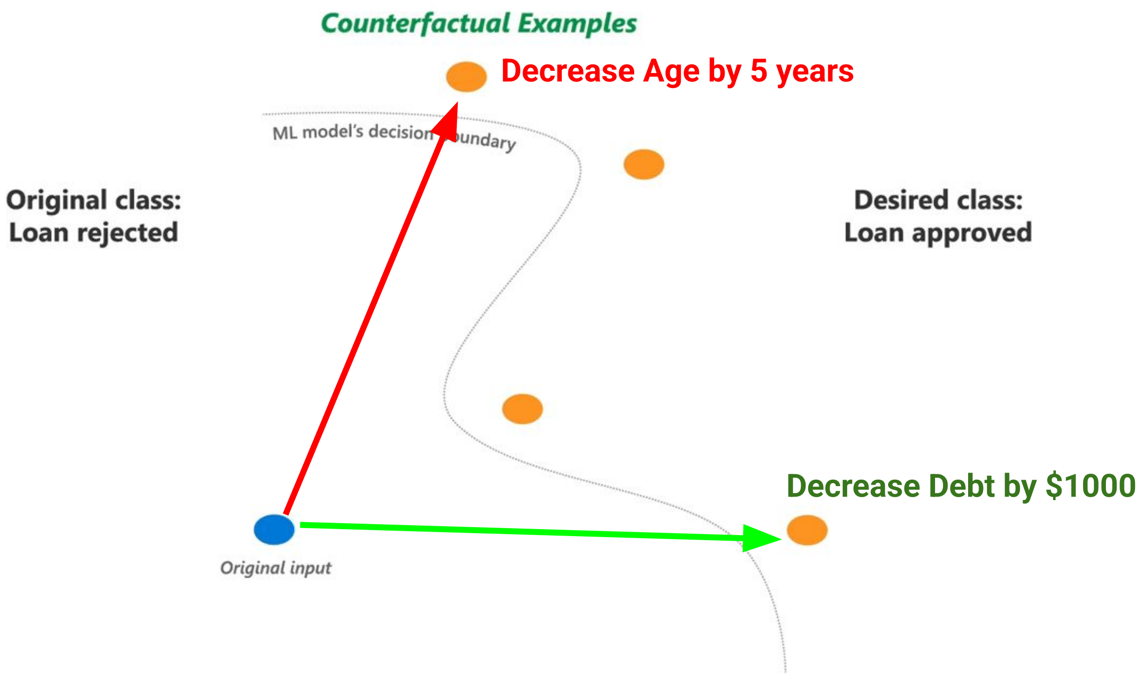
A counterfactual $\mathbf{x}'$ is valid iff it actually changes the classification outcome with respect to the original one.

$$\mathrm{Validity}(\boldsymbol{x}') = \begin{cases} 1 & \text{if } f(\boldsymbol{x}') = y_{\mathrm{target}} \\ 0 & \text{otherwise} \end{cases}$$

Goal: Hard constraint - counterfactual must be valid to be useful.

Desired Properties: **Actionability**

Only mutable features should be modified; immutable features (age, race) must remain unchanged.



Metric: **Actionability**

Count of immutable features changed:

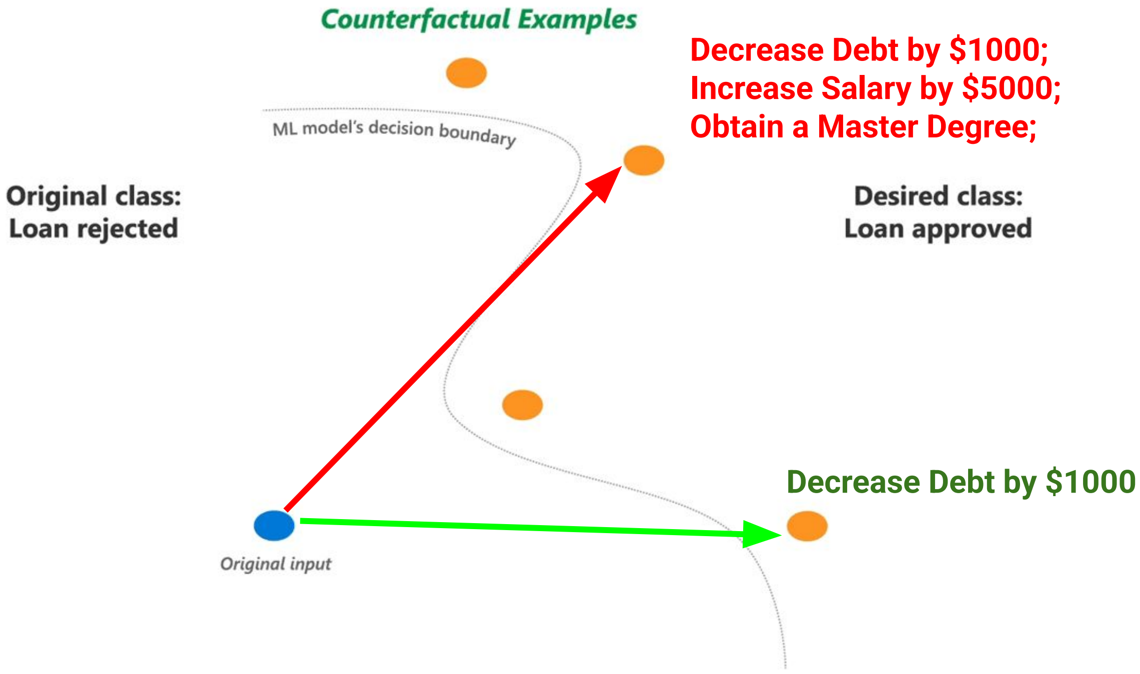
$$L_{\text{action}}(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i \in \mathcal{I}} \not\Vdash[x_i \neq x_i']$$

where $\mathcal{I}$ is the set of immutable feature indices and $\not\Vdash[\cdot]$ is the indicator function.

Goal: Minimize this metric to 0 (ideal: no immutable features changed).

Desired Properties: **Sparsity**

Prefer counterfactuals that change fewer features (simpler explanations).



Metric: **Sparsity**

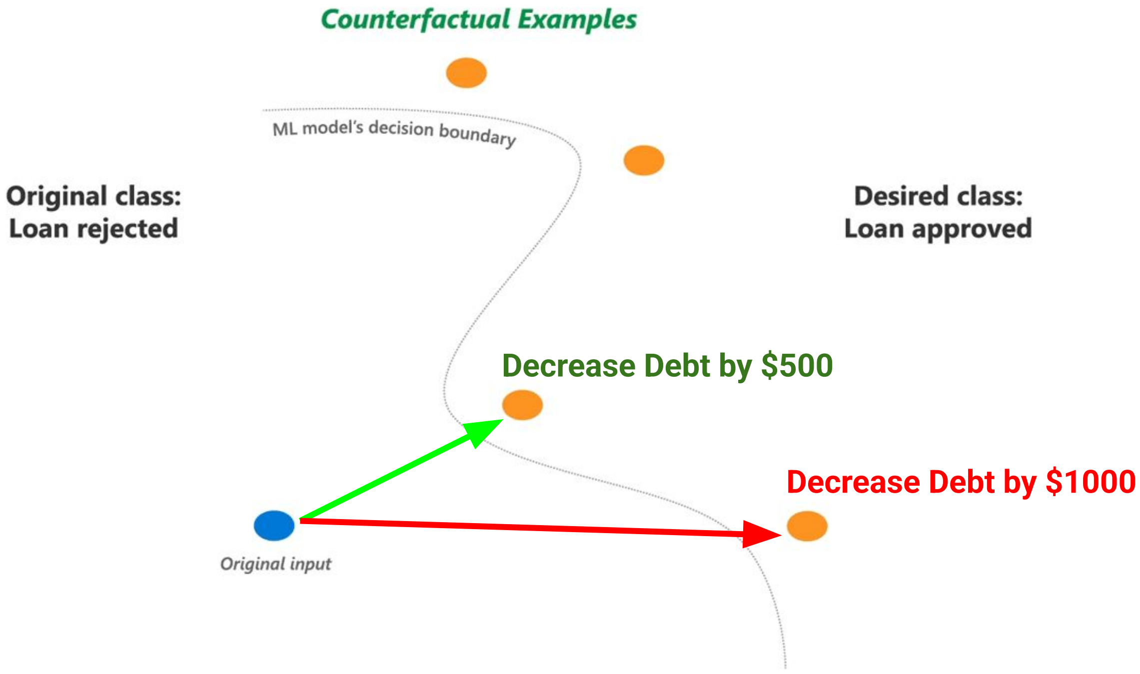
L0 Norm (count of changed features):

$$L_{\text{sparse}}(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|_0 = \sum_{i=1}^{d} \mathbb{1}[x_i \neq x_i']$$

Goal: Less changes lead to easier implementation of counterfactual explanation.

Desired Properties: **Similarity (Proximity)**

Keep the counterfactual close to the original input



Continuous Proximity vs Categorical Proximity

Euclidean Distance (L2):

$$L_{\text{prox}}^{\text{cont}}(\boldsymbol{x}, \boldsymbol{x}') = \sqrt{\sum_{i=1}^{d_c} (x_i - x_i')^2}$$

Manhattan Distance (L1):

$$L_{\text{prox}}^{\text{cont}}(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i=1}^{d_c} |x_i - x_i'|$$

Weighted Distance (normalized):

$$L_{\text{prox}}^{\text{cont}}(\boldsymbol{x}, \boldsymbol{x}') = \sqrt{\sum_{i=1}^{d_c} \frac{(x_i - x_i')^2}{\sigma_i^2}}$$

Simple Mismatch (Hamming Distance):

$$L_{\text{prox}}^{\text{cat}}(\boldsymbol{x}, \boldsymbol{x}') = \sum_{i}^{d_{cat}} \mathbb{1}[x_i \neq x_i']$$

Similarity (Proximity) - Combined Distance

Gower Distance (normalized):

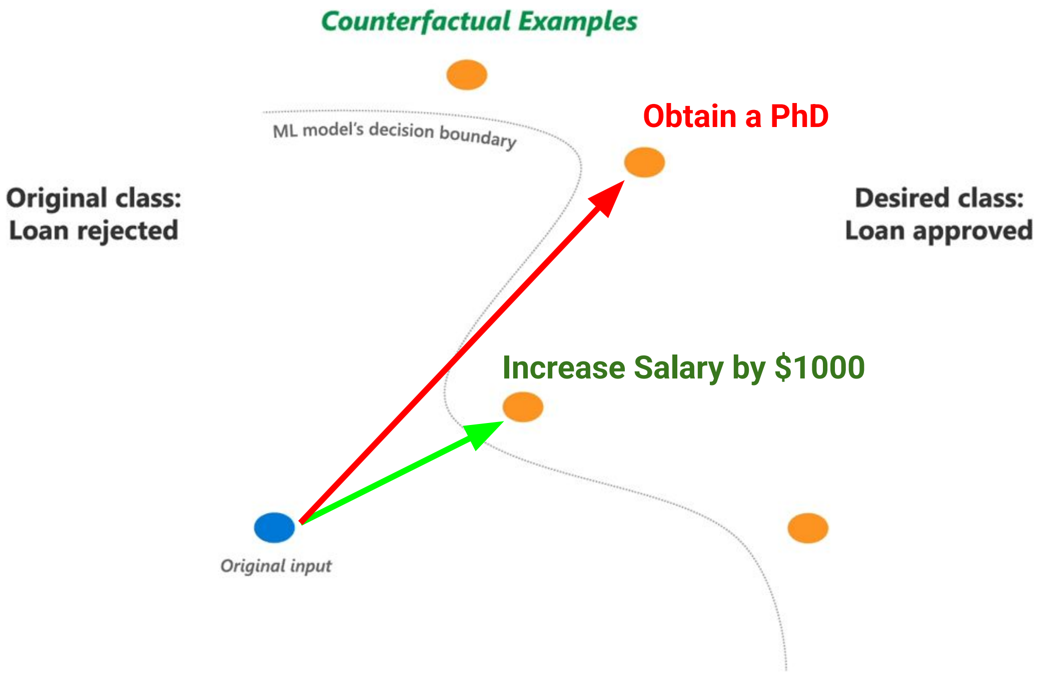
$$L_{\text{prox}}(\boldsymbol{x}, \boldsymbol{x}') = \frac{1}{d}\left(\sum_{i}^{d_{con}} \frac{|x_i - x_i'|}{R_i} + \sum_{i}^{d_{cat}} \mathbb{1}[x_i \neq x_i']\right)$$

Weighted Mixed Distance:

$$L_{\text{prox}}(\boldsymbol{x}, \boldsymbol{x}') = \alpha \cdot L_{\text{prox}}^{\text{cont}}(\boldsymbol{x}, \boldsymbol{x}') + \beta \cdot L_{\text{prox}}^{\text{cat}}(\boldsymbol{x}, \boldsymbol{x}')$$

Desired Properties: **Plausibility**

The counterfactual should lie in high-density regions of the data distribution.
Ensuring the counterfactual explanation is realistic.



Plausibility

Distance to Training Data:

$$L_{\text{plaus}}^{\text{dist}}(\boldsymbol{x}') = \min_{\boldsymbol{x}_i \in \mathcal{D}} \|\boldsymbol{x}' - \boldsymbol{x}_i\|_2$$

Likelihood-based:

$$L_{\text{plaus}}^{\text{prob}}(\boldsymbol{x}') = -\log p(\boldsymbol{x}')$$

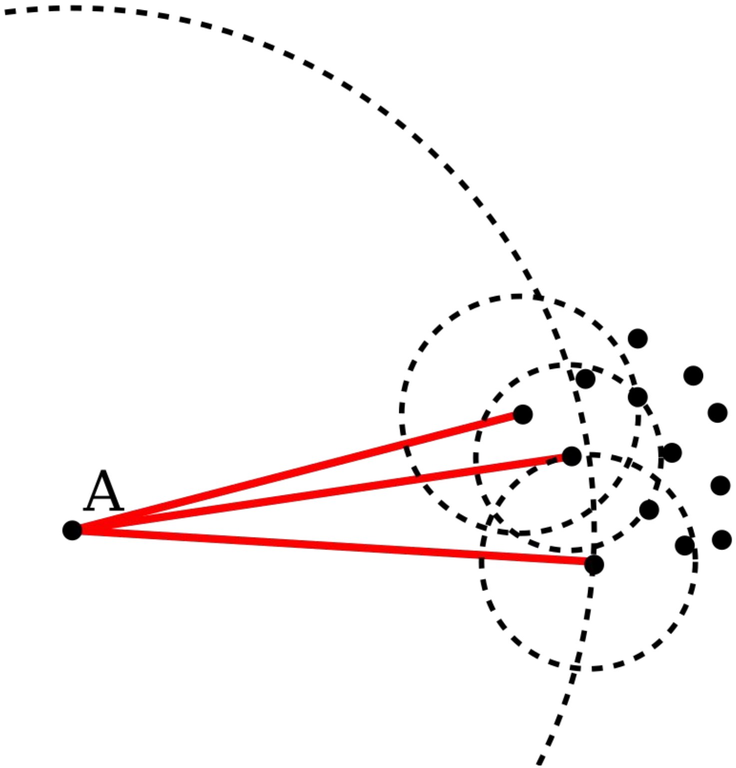
where p(**x′**) is the probability density of x′ under the data distribution.

Plausibility

Local Outlier Factor (LOF):

Basic idea: comparing the local density of a point with the densities of its neighbors.

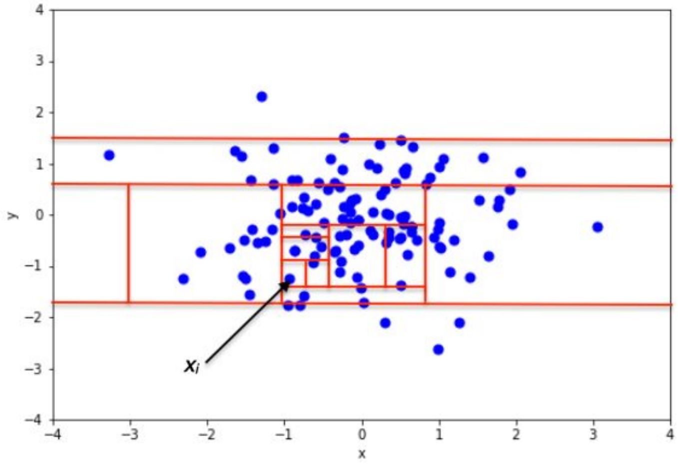
A has a much lower density than its neighbors.



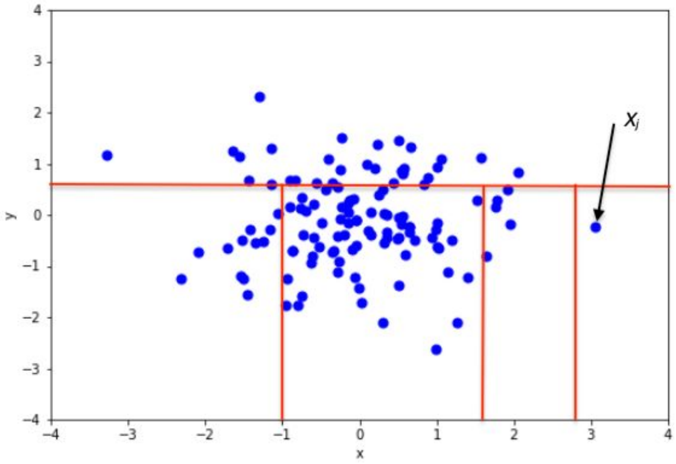
Plausibility

Isolation Forest (iForest): Measures how easily a point can be isolated using random partitions.

Plausible

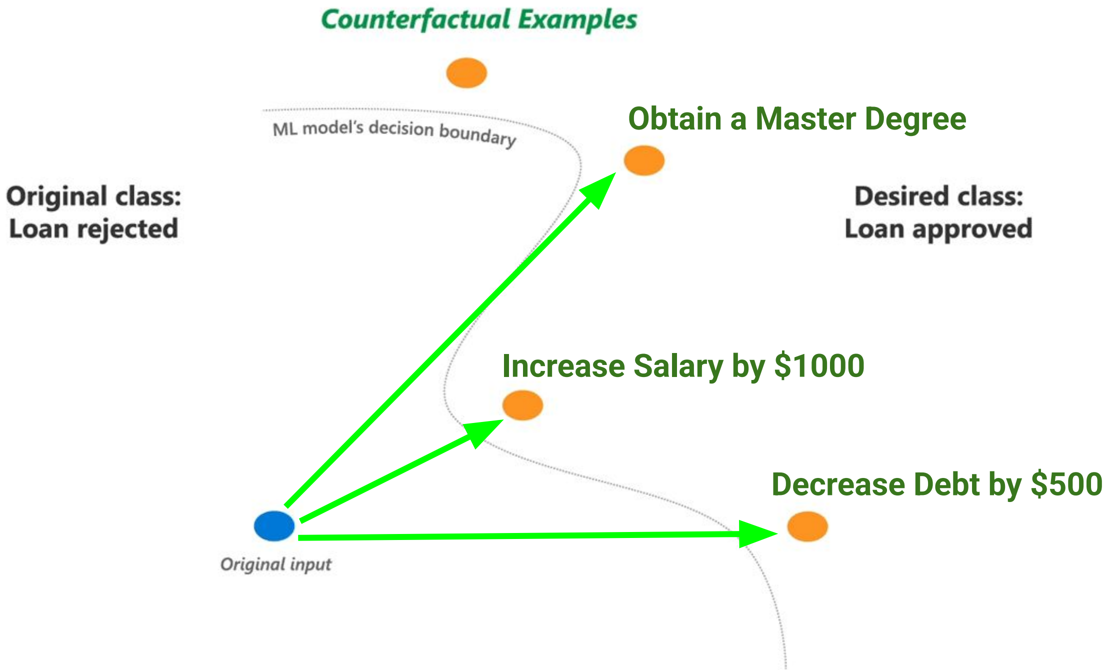


Implausible



Desired Properties: **Diversity**

Multiple diverse counterfactuals to provide users with different options.



Mothilal, Ramaravind K. et al. "Explaining machine learning classifiers through diverse counterfactual explanations." FAccT. 2020.

Diversity

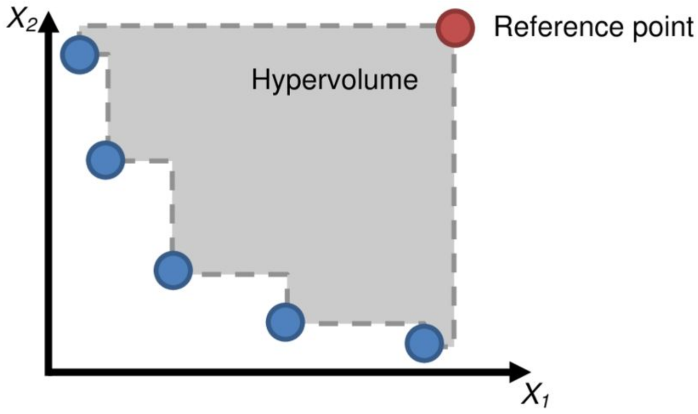
Pairwise Diversity:

For a set of counterfactuals $\{\boldsymbol{x}'_1, \boldsymbol{x}'_2, \ldots, \boldsymbol{x}'_k\}$

$$L_{\text{div}} = -\frac{1}{k(k-1)} \sum_{i=1}^{k} \sum_{j \neq i} \|\boldsymbol{x}'_i - \boldsymbol{x}'_j\|_2$$

Hypervolume Indicator:

Measures the volume of the objective space dominated by counterfactual explanations, relative to a original point.



Summary

Property	Objective
Validity	$f(\boldsymbol{x}') = y_{\text{target}}$
Actionability	Minimize $\sum_{i \in \mathcal{I}} \mathbb{1}[x_i \neq x_i']$
Sparsity	Minimize $\|\boldsymbol{x} - \boldsymbol{x}'\|_0$
Proximity (cont.)	Minimize $\sqrt{\sum_{i \in \mathcal{C}} (x_i - x_i')^2}$
Proximity (cat.)	Minimize $\sum_{i \in \mathcal{K}} \mathbb{1}[x_i \neq x_i']$
Proximity (mixed)	Minimize $\alpha \boldsymbol{L}^{\text{cont}} + \beta \boldsymbol{L}^{\text{cat}}$
Plausibility	Maximize $p(\boldsymbol{x}')$ or Minimize LOF/iForest score
Diversity	Maximize $\sum_{i \neq j} \|\boldsymbol{x}_i' - \boldsymbol{x}_j'\|$

Trade-offs Between Properties

- **Proximity vs. Sparsity:** Small changes may require modifying many features.
- **Plausibility vs. Proximity:** Closest counterfactual may be unrealistic.
- **Diversity vs. Proximity**: Diverse counterfactuals may be far from original.
- **Actionability vs. Validity:** Limiting mutable features may prevent valid counterfactuals.

💡**Key Insight:** No single ”perfect” counterfactual exists; must balance multiple competing objectives.

Multi-Objective Optimization

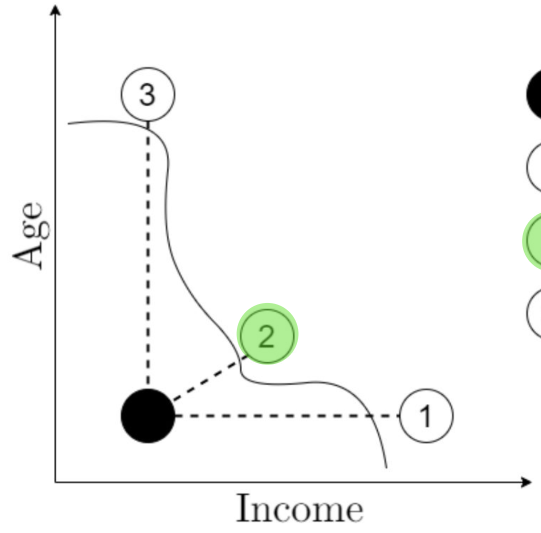
Most counterfactual methods optimize a weighted combination:

$$
\begin{aligned}
\boldsymbol{x}^* = \arg\min_{\boldsymbol{x}'} \quad & \lambda_1 L_{\text{validity}}(\boldsymbol{x}') \\
& + \lambda_2 L_{\text{prox}}(\boldsymbol{x}, \boldsymbol{x}') \\
& + \lambda_3 L_{\text{sparse}}(\boldsymbol{x}, \boldsymbol{x}') \\
& + \lambda_4 L_{\text{plaus}}(\boldsymbol{x}') \\
& + \lambda_5 L_{\text{action}}(\boldsymbol{x}, \boldsymbol{x}')
\end{aligned}
$$

where λ_i are hyperparameters controlling the trade-offs.

Endogenous vs Exogenous Counterfactual Explanations

Endogenous counterfactuals are crafted using feature values from existing data instances

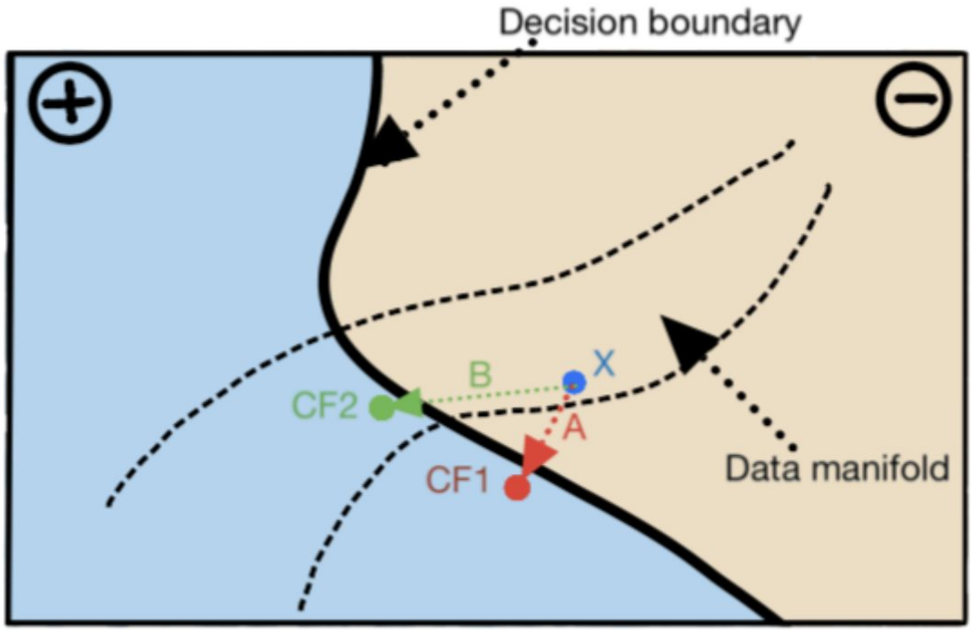


	Income	Age
●	$ 32,000	39
1	$ 40,000	39
2	$ 36,000	44
3	$ 32,000	140

Figure 1: Example of counterfactual explanations for loan approval.

Endogenous vs Exogenous Counterfactual Explanations

Exogenous counterfactuals: generated through interpolations or random data generation, do not strictly rely on existing dataset features.



Method by Wachter et al. (2017)

Key Idea: Find closest counterfactual by minimizing weighted loss combining prediction loss and distance.

$$\mathbf{x}' = \arg\min_{\mathbf{x}'} \boxed{\lambda} \cdot \boxed{\ell(f(\mathbf{x}'), y_{\text{target}})} + \boxed{\|\mathbf{x} - \mathbf{x}'\|_1}$$

λ controls trade-off between validity and proximity

Prediction loss (e.g., cross-entropy, hinge loss)

Distance between original input and CF

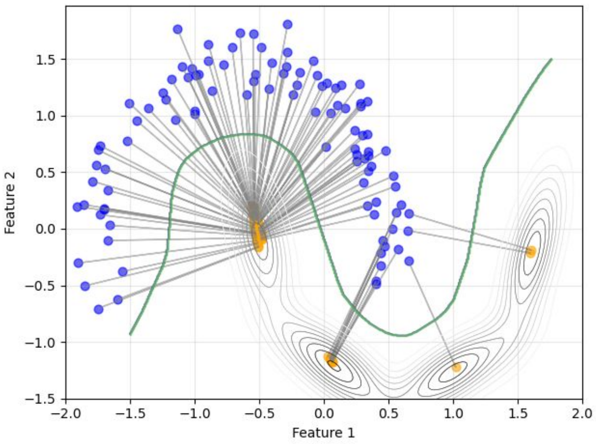
Advantages:
- Simple formulation
- Model-agnostic
- Possible gradient-based optimization
- Promotes sparsity via L1 norm

Disadvantages:
- Single counterfactual (no diversity)
- No plausibility
- May produce unrealistic counterfactuals
- No guarantee on number of features changed

Method by Artelt et al. (2020)

Key Idea: Incorporate plausibility through probability distributions, ensuring counterfactuals lie in high-density regions.

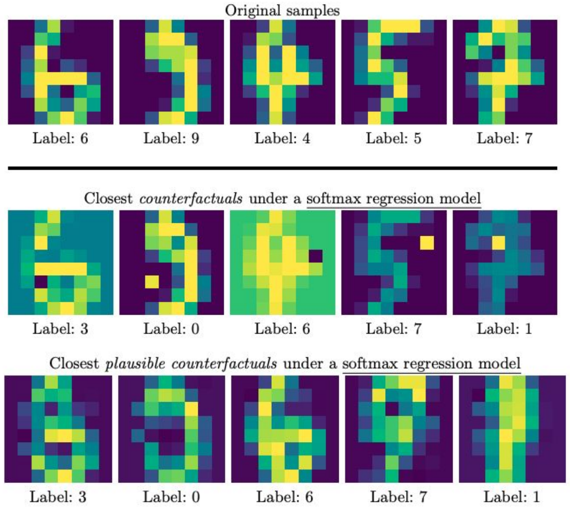
Objective Function: $$\mathbf{x}' = \arg\min_{\mathbf{x}'} \|\mathbf{x} - \mathbf{x}'\|_2^2$$
$$\text{s.t.} \quad f(\mathbf{x}') \neq f(\mathbf{x}) \quad \text{(validity)}$$
$$p(\mathbf{x}') \geq \epsilon \quad \text{(plausibility)}$$



Key Innovation: Convex Approximation

- Use Gaussian Mixture Model (GMM) for density estimation
- Approximate GMM density: $\hat{p}(x) = \max_j\{\pi_j \mathcal{N}(x|\mu_j, \Sigma_j)\}$
- Converts to convex quadratic constraints:

$$(x - \mu_j)^\top \Sigma_j^{-1}(x - \mu_j) + c_j \leq \delta'$$



Practical algorithm:

1. Try each Gaussian component j = 1, 2, ..., m separately
2. For each j, solve a convex optimization problem
3. Pick the solution with the smallest distance to x

Constrained optimization

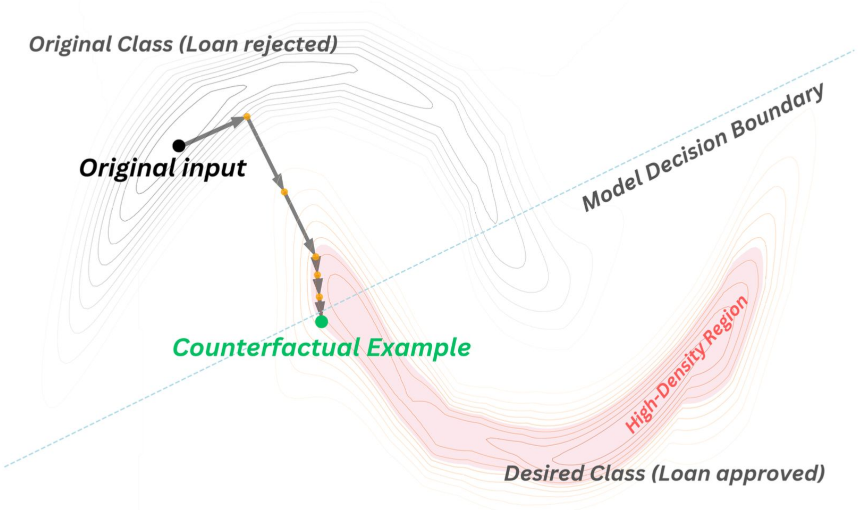
PPCEF: Probabilistically Plausible Counterfactual Explanations with Normalizing Flows

Key Idea: Combine normalizing flows for conditional density estimation p(x|y) with unconstrained optimization to ensure both validity and probabilistic plausibility.

$$\boldsymbol{x}^* = \arg\min_{\boldsymbol{x}'} d(\boldsymbol{x}_0, \boldsymbol{x}') + \lambda \cdot (\ell_v(\boldsymbol{x}', y') + \ell_p(\boldsymbol{x}', y'))$$

Validity Loss (ℓ_v): $\boxed{\ell_v(x', y')} = \max(0.5 + \epsilon - p_d(y'|x'), 0)$

Plausibility Loss (ℓ_p): $\boxed{\ell_p(x', y')} = \max(\delta - p(x'|y'), 0)$



Guaranteed plausibility: Always produces 100% probabilistically plausible CFs
High-dimensional data: Works well with complex, high-dim distributions
Unconstrained formulation: Simple gradient-based optimization
Batch processing: Orders of magnitude faster than alternatives
No parametric assumptions: Flexible density modeling

Wielopolski et al. Probabilistically Plausible Counterfactual Explanations with Normalizing Flows. ECAI 2024

DiCE: Diverse Counterfactual Explanations

Key Idea: Generate a set of k diverse counterfactuals providing users with multiple actionable options.

Optimization: Genetic algorithm or gradient-based.

$$\{\mathbf{x}'_1, \ldots, \mathbf{x}'_k\} = \underset{\{\mathbf{x}'_i\}}{\arg\min} \; \frac{1}{k}\sum_{i=1}^{k}\Big[\text{yloss}(f(\mathbf{x}'_i), y_{\text{target}}) + \lambda_1 \cdot \text{dist}(\mathbf{x}, \mathbf{x}'_i) + \lambda_2 \cdot \text{sparse}(\mathbf{x}, \mathbf{x}'_i)\Big] - \lambda_3 \cdot \text{dpp\_diversity}(\{\mathbf{x}'_i\})$$

```
Query instance (original outcome : 0)
```

	age	workclass	education	marital_status	occupation	race	gender	hours_per_week	income
0	22.0	Private	HS-grad	Single	Service	White	Female	45.0	0.01904

```
Diverse Counterfactual set (new outcome : 1)
```

	age	workclass	education	marital_status	occupation	race	gender	hours_per_week	income
0	70.0	-	Masters	-	White-Collar	-	-	51.0	0.534
1	-	Self-Employed	Doctorate	Married	-	-	-	-	0.861
2	47.0	-	-	Married	-	-	-	-	0.589
3	36.0	-	Prof-school	Married	-	-	-	62.0	0.937

$$\text{dpp\_diversity}(\{\mathbf{x}'_i\}) = \det\left(\exp\left(-\frac{\|\mathbf{x}'_i - \mathbf{x}'_j\|_2^2}{2\sigma^2}\right)\right)$$

Multiple diverse counterfactuals (user choice)
Supports various distance metrics
Handles mixed feature types

Computationally expensive
Hyperparameter tuning required (λ values, k)
DPP computation scales poorly with k
May generate implausible counterfactuals

Mothilal, Ramaravind K. et al. "Explaining machine learning classifiers through diverse counterfactual explanations." FAccT 2020.

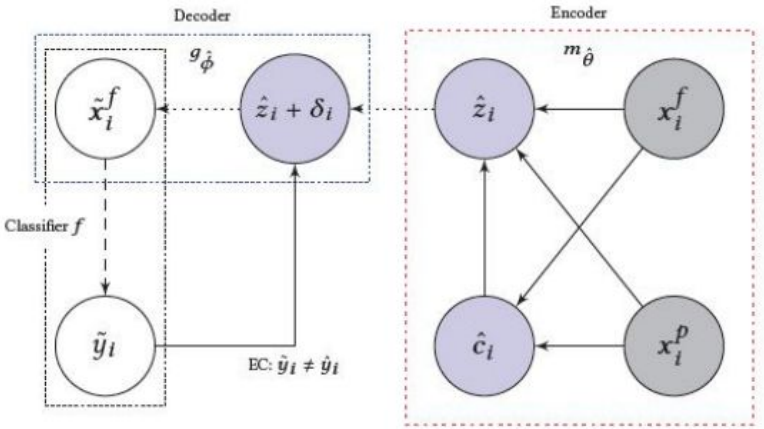
C-CHVAE: Conditional Heterogeneous Variational Autoencoder

Key Idea: Generate counterfactuals that flip predictions while staying on the data manifold.

Encoder: $z = m_\theta(x^f | x^p)$

Mixture Prior: $p(z | c, x^p)$

Decoder: $\tilde{x}^f = g_\phi(z | x^p)$



Search Procedure:

1. **Encode** original input: $\hat{z} = m_\theta(x^f | x^p)$
2. **Sample** perturbations uniformly on sphere: $\tilde{z} \sim \mathcal{S}(\hat{z}, r)$
3. **Decode** candidate: $\tilde{x}^f = g_\phi(\tilde{z} | x^p)$
4. **Test** prediction: if $f(\tilde{x}^f, x^p) \neq f(x^f, x^p) \rightarrow$ found!
5. **Expand** search radius r if not found, repeat

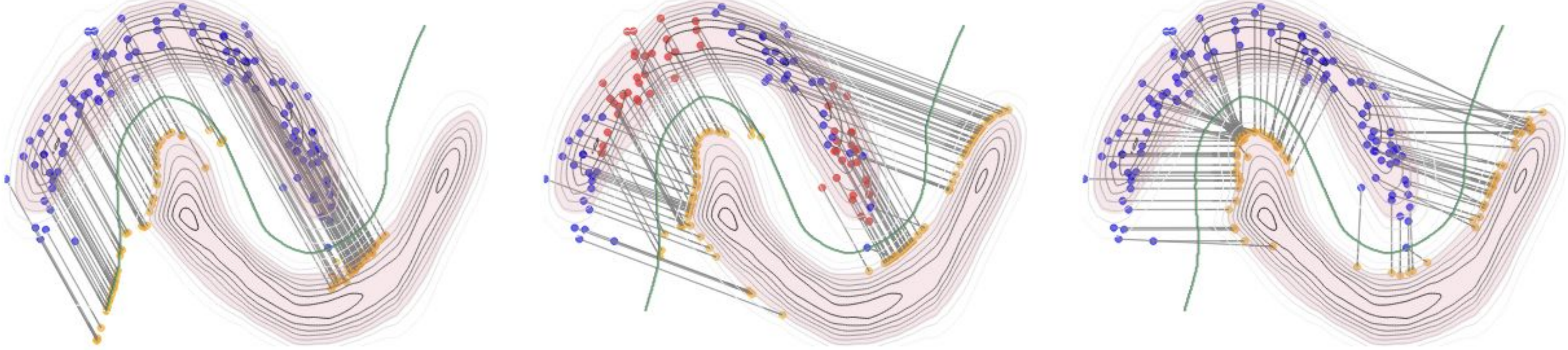
Inherent **plausibility**
Fast generation after training
Generate **diverse** counterfactuals via sampling

Requires training generative model
VAE reconstruction quality affects results

Pawelczyk, M. et al. "Learning model-agnostic counterfactual explanations for tabular data." WWWC 2020

Global, **Group-Wise** and **Local** Counterfactual Explanations

Shift-based Representation: $\mathbf{x}' = \mathbf{x}_0 + \mathbf{d}$, where shift vector $\mathbf{d} \in \mathbb{R}^D$ applicable to all instances



(a) Global CFs — **Single** change vector **d**

(b) Group-wise CFs — **Several** change vectors **d**

(c) Local CFs — **Unique** change vector **d** per data point

Global, **Group-Wise** and **Local** Counterfactual Explanations

Shift-based Representation: $\mathbf{x}' = \mathbf{x}_0 + \mathbf{d}$, where shift vector $\mathbf{d} \in \mathbb{R}^D$ applicable to all instances

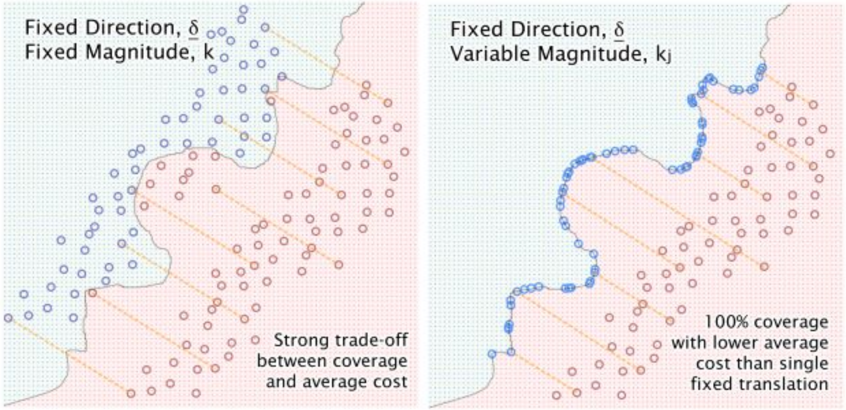
Example of Global Counterfactual Explanations:
Instead of:
- Alice: Income +$10k, Debt -$2k
- Bob: Income +$8k, Education: Master
- Carol: Income +$12k

Find ONE for ALL:
- ”Increase income +$12k, reduce debt -$6k”

GLOBE-CE: Global & Efficient Counterfactual Explanations

Key Idea: Fixed direction + variable magnitude = interpretable + efficient

Variable magnitudes per instance: $\mathbf{x}' = \mathbf{x}_0 + k \cdot \mathbf{d}$



Proposed Approach - Random Sampling:

1. Sample n_s random change vectors at fixed cost c
2. For each hange vector:
   - Apply to all instances
   - Calculate validity
3. Greedily select n change vectors with highest validity
4. Ensure diversity (new direction must be sufficiently different)

Very fast - seconds instead of hours
Simple - easy to understand and implement
Model-agnostic - works with any black-box model
Interpretable - clear rules and profiles

Plausibility not guaranteed
Random sampling may miss optimal directions
Simple cost model (L0 Distance)

Ley, D. et al. "GLOBE-CE: A translation based approach for global counterfactual explanations." ICML 2023.

GLOBE-CE: Global & Efficient Counterfactual Explanations

Example Counterfactual Explanations:

If **ForeignWorker** = **False**:
NumInqLast6M: -0.77 [0.12]
NumInqLast6Mexcl7days: +0.66 [0.10]
NetFractionRevolvingBurden: -0.30 [0.02]
[**Mean Cost = 0.155**]

If **ForeignWorker** = **True**:
NumInqLast6M: -2.64 [0.40]
NumInqLast6Mexcl7Days: +2.26 [0.34]
NetFractionRevolvingBurden: -1.01 [0.06]
[**Mean Cost = 0.531**]

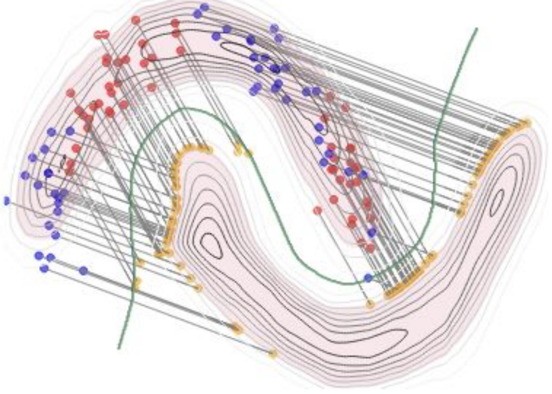
Unifying Perspectives: Plausible Counterfactual Explanations on Global, Group-wise, and Local Levels

Key Idea: Simultaneous group discovery and CF generation for groups

Formulation: $\mathbf{X}'_{GW} = \mathbf{X}_0 + \mathbf{KSD}_{GW}$

where:

- $\mathbf{D}_{GW} \in \mathbb{R}^{K \times D}$: K base shifting vectors $[\mathbf{d}_1, \ldots, \mathbf{d}_K]^T$
- $\mathbf{S} \in [0,1]^{N \times K}$: Selection matrix (soft assignment)
- $\mathbf{K} = \mathrm{diag}(k_1, \ldots, k_N)$: Instance-specific magnitudes



(b) Group-wise CFs

To Get	Set Number of Groups	Result
Local	$K = N$ (one per person)	Individual explanations
Global	$K = 1$ (one for all)	Single rule for everyone
Group-wise	$K =$ automatic	Learns optimal groups

Unifying Perspectives: Plausible Counterfactual Explanations on Global, Group-wise, and Local Levels

$$\arg\min_{\mathbf{K},\mathbf{B},\mathbf{D}_{GW}} d^G(\mathbf{X}_0, \mathbf{X}'_{GW}) + \omega \cdot \ell_v(h(\mathbf{X}'_{GW}), y') + \omega_p \cdot \ell_p(\mathbf{X}'_{GW}, y')$$

$$+\omega_s \cdot \ell_s(\mathbf{B}) + \omega_k \cdot \ell_k(\mathbf{B}) + \omega_d \cdot \ell_d(\mathbf{D}_{GW})$$

Regularization Terms

- $\ell_s(\mathbf{B})$: Entropy regularization for sparse assignment to groups
- $\ell_k(\mathbf{B})$: Group number reduction (automatic detection)
- $\ell_d(\mathbf{D}_{GW})$: Diversity regularization using determinant
- $\ell_p(\mathbf{X}'_{GW}, y')$: Plausibility constraint via normalizing flows

Unifying Perspectives: Plausible Counterfactual Explanations on Global, Group-wise, and Local Levels

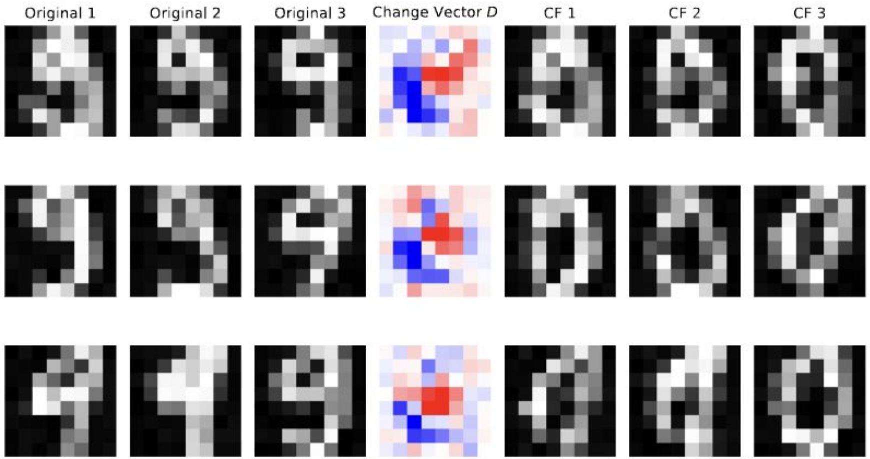
Unified - one method for all three levels
Automatic grouping - finds optimal number
Plausible - ensures realistic suggestions
End-to-end - no separate clustering
Best quality - superior validity and plausibility

Requires differentiable models
Slower than GLOBE-CE
More complex to implement
More hyperparameters to tune

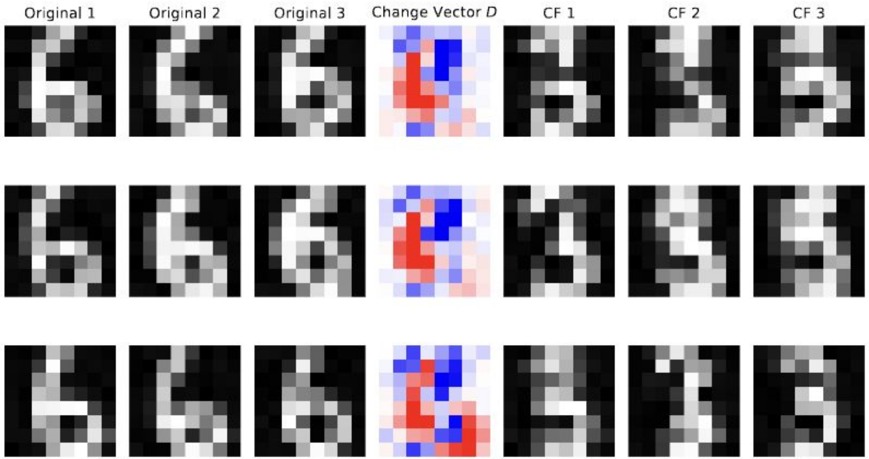
Unifying Perspectives: Plausible Counterfactual Explanations on Global, Group-wise, and Local Levels

Example Counterfactual Explanations:

Unifying Perspectives: Plausible Counterfactual Explanations on Global, Group-wise, and Local Levels



(a) Counterfactual explanations for class 9 with the desired class 0.



(b) Counterfactual explanation for class 6 with desired class 3.

When to use what?

Use **Local CFs** when you need to:

Personalized Explanations

- Explain individual decisions to specific people
- Provide tailored, actionable advice
- Handle unique or edge cases

Best Scenarios:

- Customer service interactions
- Medical diagnosis for specific patient
- Individual loan applications
- Legal right-to-explanation requests
- One-on-one counseling

Avoid When:

- Need to understand overall model behavior
- Looking for systematic biases
- Have thousands of cases to explain
- Want to compare across groups
- Need policy-level insights

When to use what?

Use **Global CFs** when you need to:

High-Level Understanding

- Understand general model behavior
- Communicate simple, universal rules
- Quick model auditing and validation
- Policy recommendations that apply broadly

Best Scenarios:

- Model documentation
- Regulatory compliance reports
- Public communication of policies
- Quick bias screening
- High-level strategic decisions

Avoid When:

- Model is highly non-linear
- Different subgroups behave very differently
- Need personalized advice
- Single rule won't fit most cases

When to use what?

Use **Group-wise CFs** when you need to:

Balanced Understanding

- Identify patterns within subpopulations
- Detect differential treatment across groups
- Provide targeted but scalable recommendations
- Balance detail with interpretability

Best Scenarios:

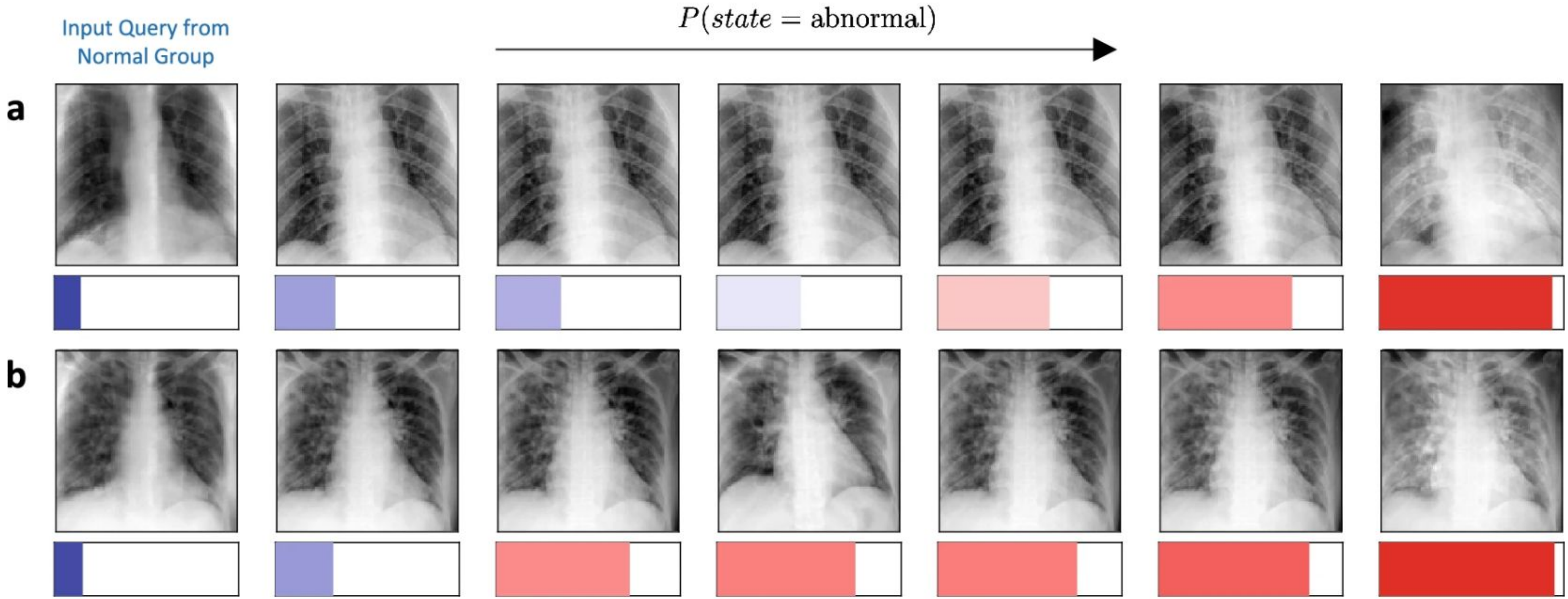
- Fairness analysis (compare by demographics)
- Market segmentation (different customer types)
- Medical guidelines (different risk groups)

Avoid When:

- Everyone truly needs different advice
- Only care about overall trend
- Need instant individual responses
- Simple global rule suffices

Let’s move to practical part!

https://github.com/LukiLenkiewicz/ML-in-PL-counterfactuals

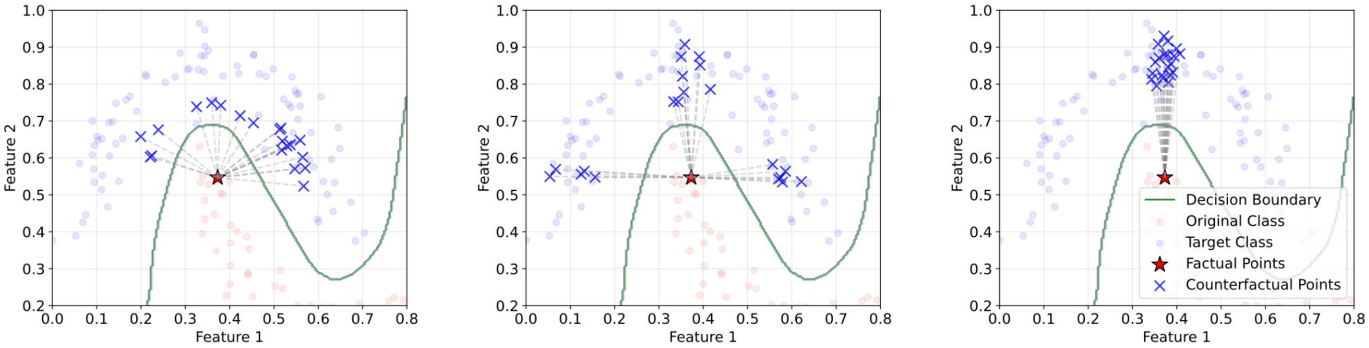


Input Query from Normal Group
$P(state = \text{abnormal})$
a
b

DiCoFlex: Model-agnostic diverse counterfactuals with flexible control

Normalizing Flow Model:

$$p_\theta(\mathbf{x}'|\mathbf{x}, y') = p_Z(f_\theta(\mathbf{x}'; \mathbf{x}, y')) \cdot |\det J_{f_\theta}(\mathbf{x}'; \mathbf{x}, y')|$$



(a) Unconstrained (b) Sparsity constraints (c) Actionability constraints